```c
//Write a menu based C program for sparse matrix realization and make a compact form of it and perform simple transpose and f

#include <stdio.h>
#define max 8
int val=1,m,n;                                          //global variables for l=no of nonzero values, m=no of rows , n=
int s[max][max],c[max][3],st[max][3],ft[max][3];
int i, j,zeros=0;

//Sparse matrix input function
void sparsematrix()
{
    printf("Enter elements of the sparse matrix:\n");  //Inputing the sparse matrix
    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &s[i][j]);
        }
        printf("\n");
    }

    printf("The sparse matix is:");                   //Displaying the sparse matrix
    printf("\n");
    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
        {
            printf("%d\t", s[i][j]);
            if (s[i][j]==0){
                zeros+=1;
            }
        }
        printf("\n");
    }
    printf("\n");
}

//Compact matrix conversion function
void compactmatrixconverion()
{
    c[0][0]=m,c[0][1]=n;
    for(i = 0; i < m; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(s[i][j]!=0)
            {
                c[val][0]=i,c[val][1]=j,c[val][2]=s[i][j];
                val+=1;
            }
        }
    }
    c[0][2]=(val-1);

    printf("The compact matix is:");                   //Displaying the compact matrix
    printf("\n");
    for(i = 0; i < val; i++)
    {
        for(j = 0; j < 3; j++)
        {
            printf("%d\t", c[i][j]);
        }
        printf("\n");


    }
    printf("\n");
}

void simpletranspose()
{
    st[0][0] = c[0][1];
    st[0][1] = c[0][0];
    st[0][2] = c[0][2];
    for (i=1; i < val; i++)
    {
        st[i][0]=c[i][1];
        st[i][1]=c[i][0];
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        st[i][2]=c[i][2];
    }

    printf("The simple transpose of compact matix is:");                    //Displaying the simple transposed matrix
    printf("\n");
    for(i = 0; i < val; i++)
    {
        for(j = 0; j < 3; j++)
        {
            printf("%d\t", st[i][j]);
        }
        printf("\n");
    }
    printf("\n");
}

void fasttranspose()
{
    ft[0][0] = c[0][1];
    ft[0][1] = c[0][0];
    ft[0][2] = c[0][2];

    int rowterms[max];
    int startpos[max];
    int k;

    for (i = 0; i < n; i++) {
        rowterms[i] = 0;
    }

    for (i = 1; i < val; i++) {
        rowterms[c[i][1]] += 1;
    }

    startpos[0] = 1;
    for (i = 1; i < n; i++) {
        startpos[i] = startpos[i-1] + rowterms[i-1];
    }

    for (i = 1; i < val; i++) {
        k = startpos[c[i][1]]++;
        ft[k][0] = c[i][1];
        ft[k][1] = c[i][0];
        ft[k][2] = c[i][2];
    }

    printf("The fast transpose of compact matix is:");
    printf("\n");
    for(i = 0; i < val; i++)
    {
        for(j = 0; j < 3; j++)
        {
            printf("%d\t", ft[i][j]);
        }
        printf("\n");
    }

}

//Main function
int main()
{
    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);
    sparsematrix();
    if (zeros==0){
        printf("Its not a sparse matrix");
    }
    else if(zeros==(m*n)){
        printf("Its a null matrix ");
    }
    else{
        compactmatrixconverion();
        simpletranspose();
        fasttranspose();
    }
    return 0;
}
```